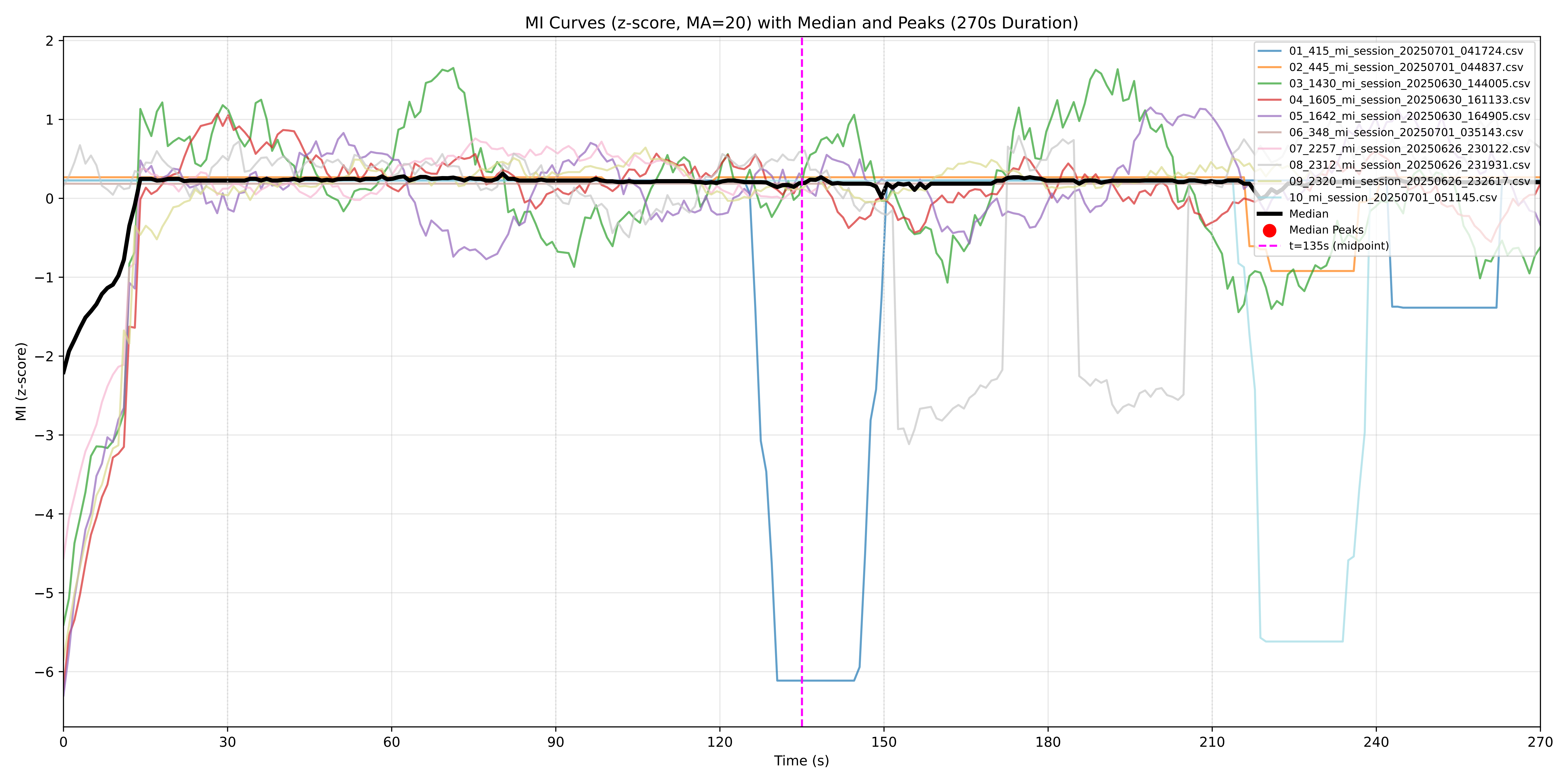
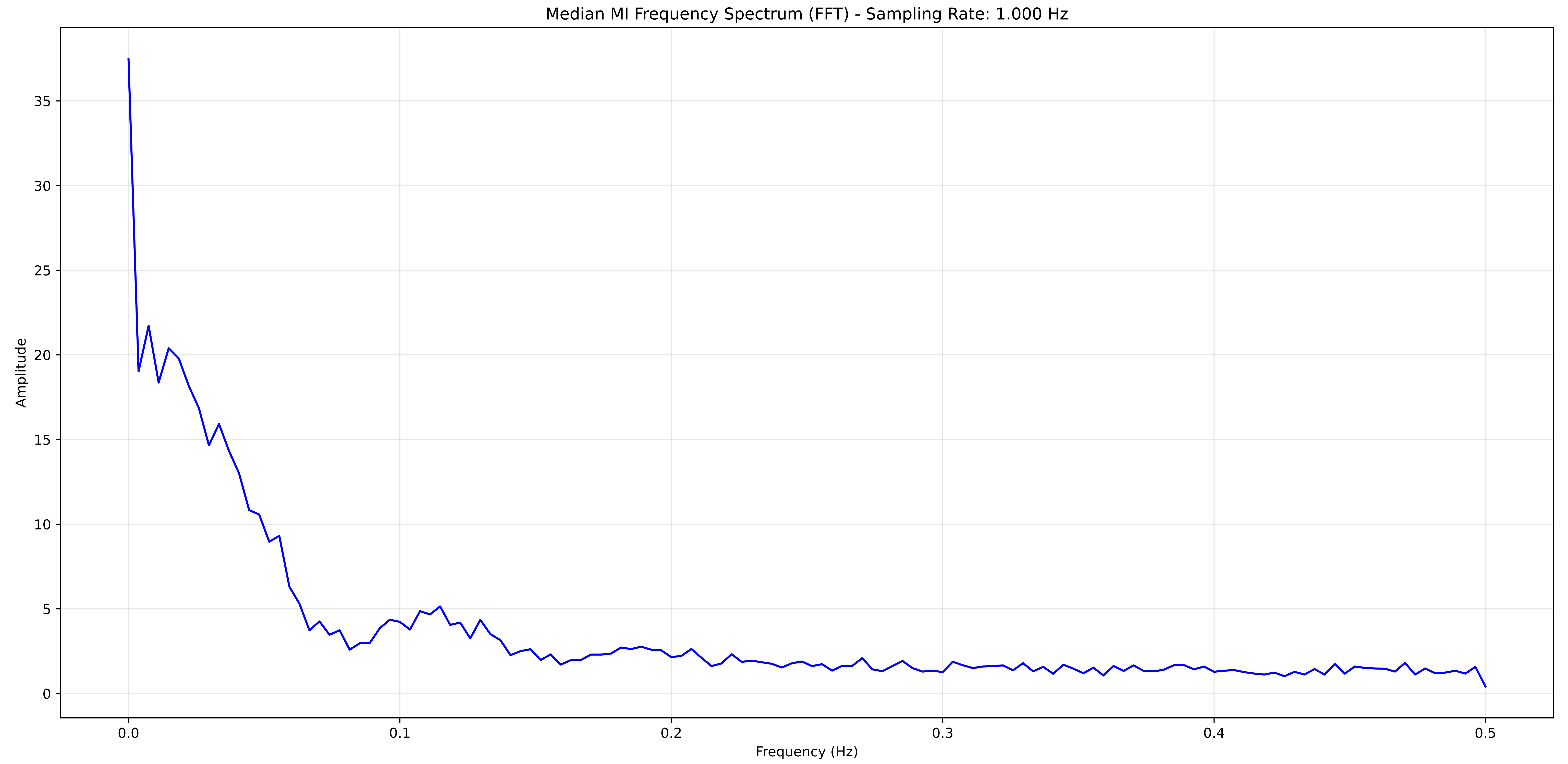
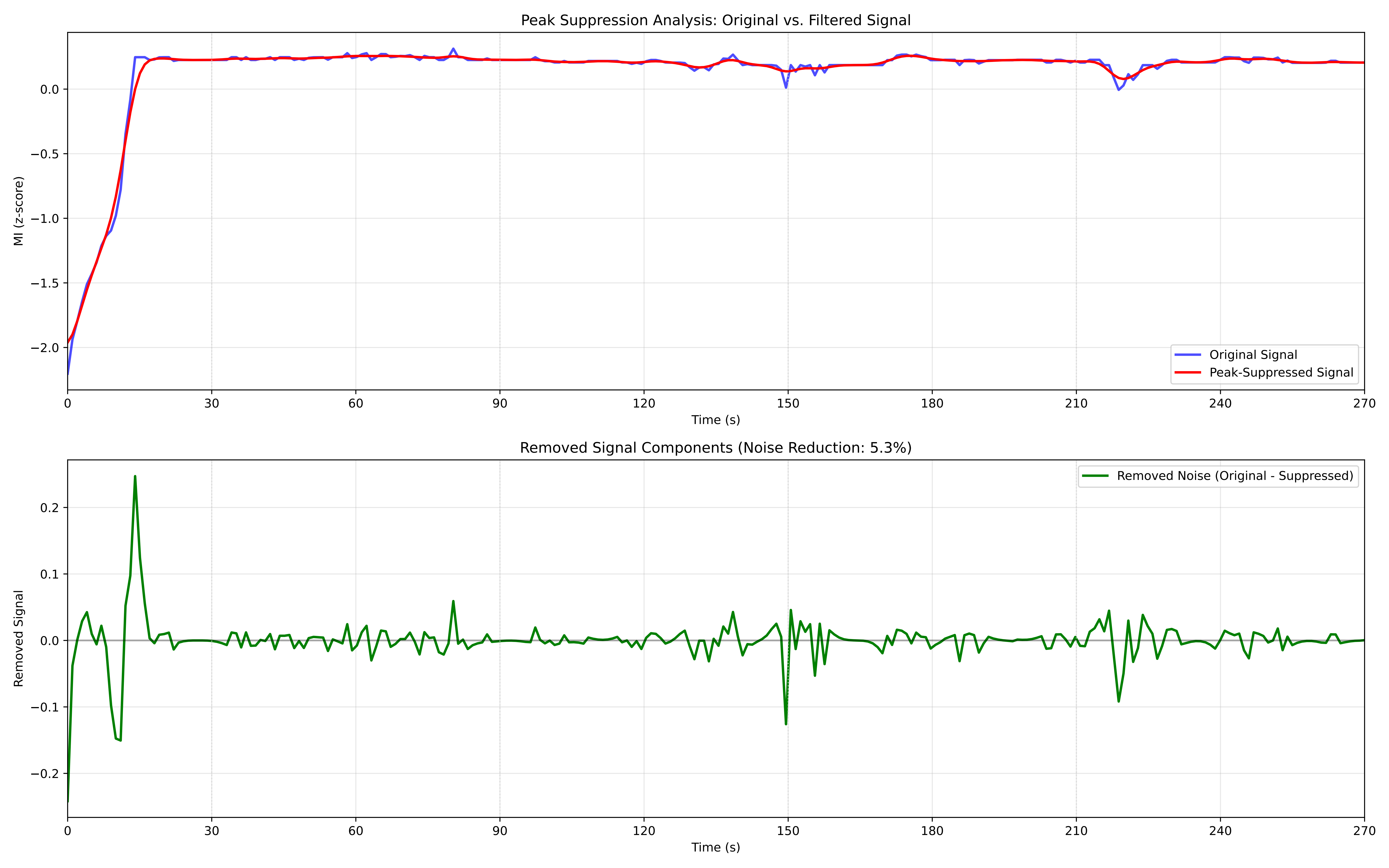
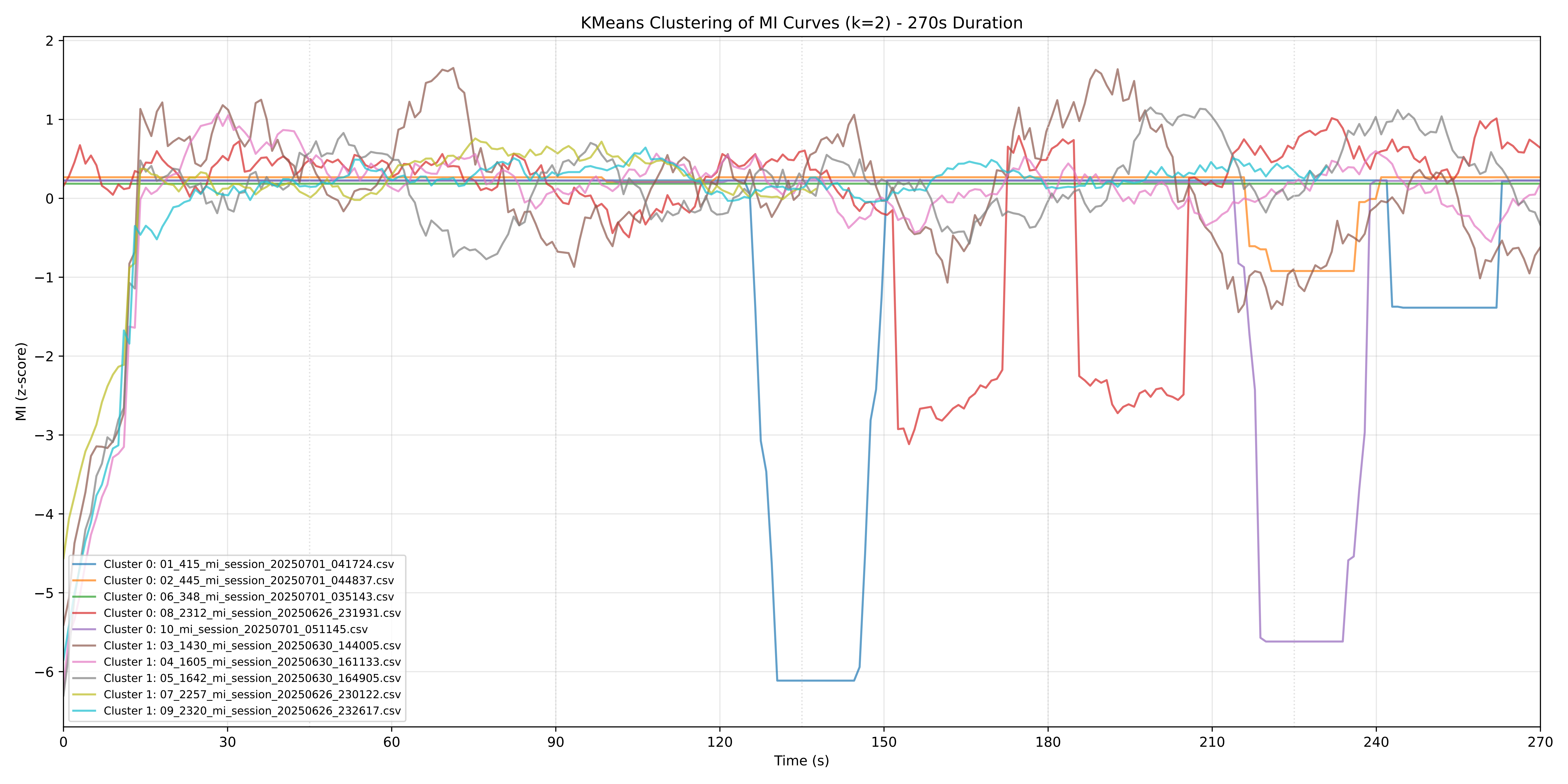
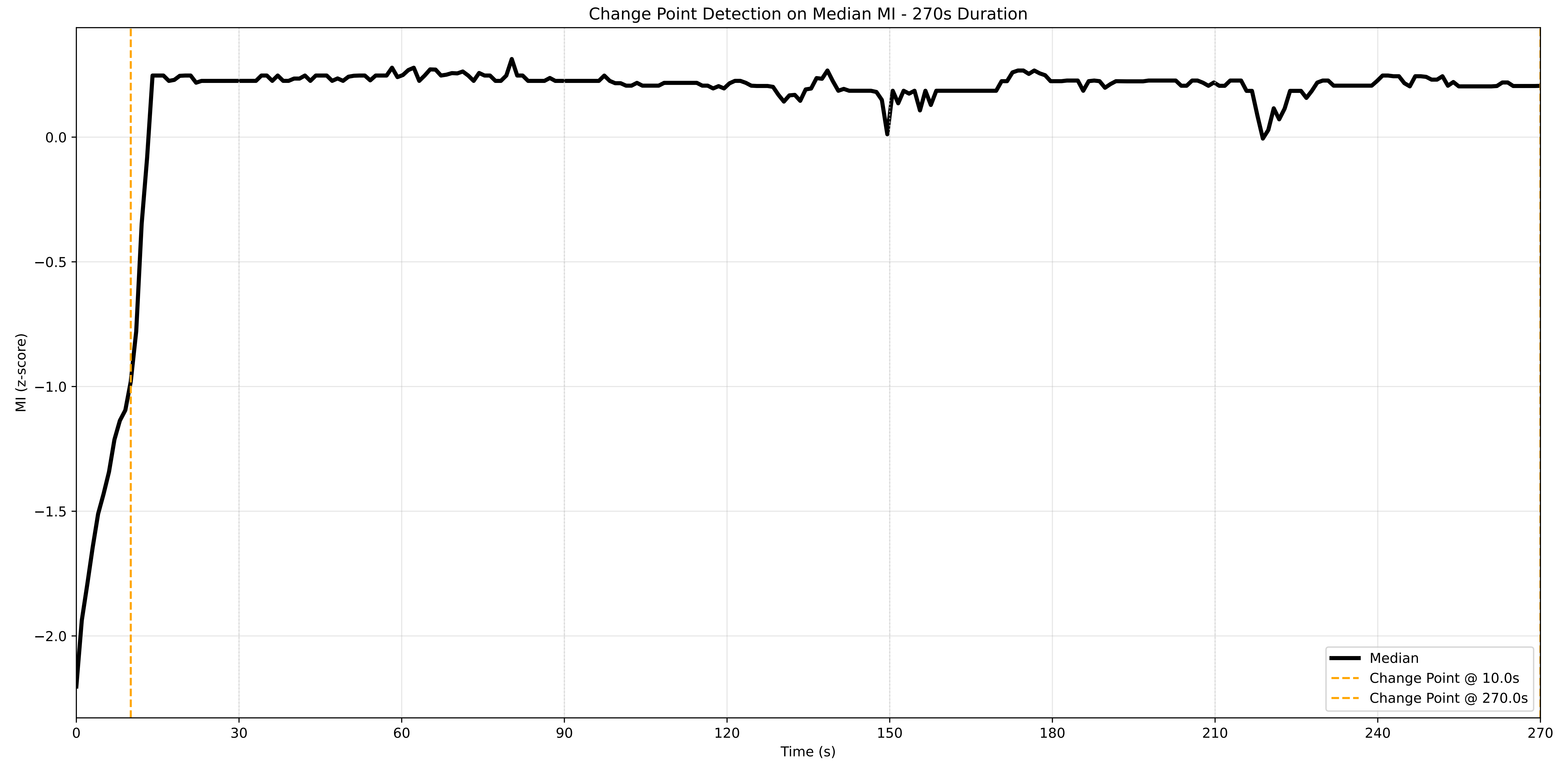

MI Curves (z-score, MA=20) with Median and Peaks (270s Duration)
01_415_mi_session_20250701_041724.csv
02_445_mi_session_20250701_044837.csv
03_1430_mi_session_20250630_144005.csv
04_1605_mi_session_20250630_161133.csv
05_1642_mi_session_20250630_164905.csv
06_348_mi_session_20250701_035143.csv
07_2257_mi_session_20250626_230122.csv
08_2312_mi_session_20250626_231931.csv
09_2320_mi_session_20250626_232617.csv
10_mi_session_20250701_051145.csv
Median
Median Peaks
t=135s (midpoint)
MI (z-score)
Time (s)
2
1
0
−1
−2
−3
−4
−5
−6
0
30
60
90
120
150
180
210
240
270



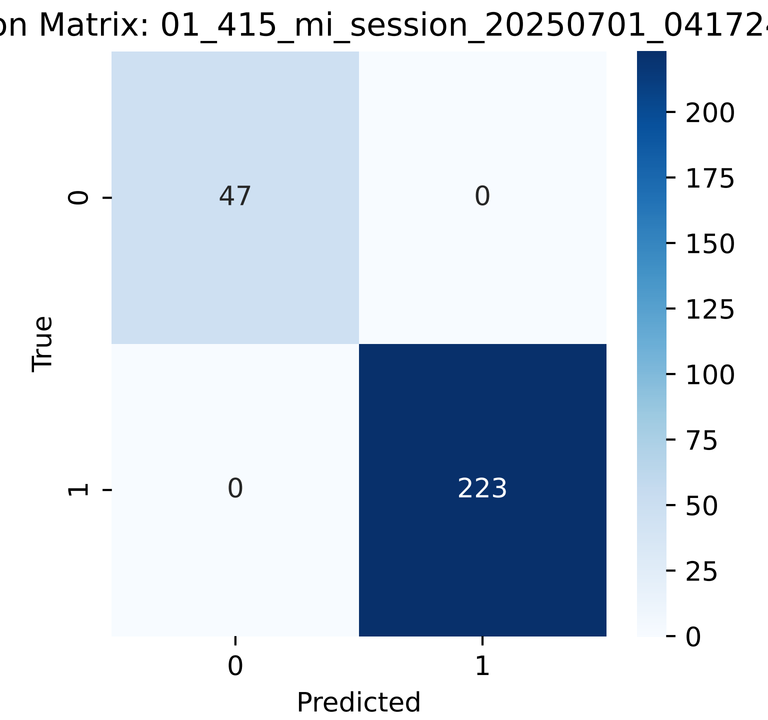
Median MI Frequency Spectrum (FFT) - Sampling Rate: 1.000 Hz
Amplitude
Frequency (Hz)
0
5
10
15
20
25
30
35
0.0
0.1
0.2
0.3
0.4
0.5


Peak Suppression Analysis: Original vs. Filtered Signal
MI (z-score)
Time (s)
Original Signal
Peak-Suppressed Signal
Removed Signal Components (Noise Reduction: 5.3%)
Removed Signal
Removed Noise (Original - Suppressed)
Time (s)

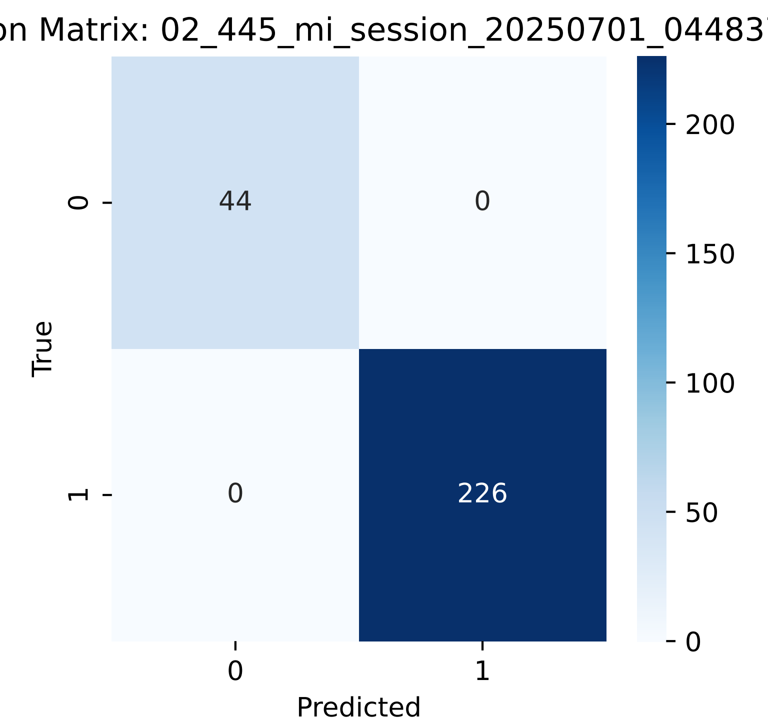

KMeans Clustering of MI Curves (k=2) - 270s Duration
MI (z-score)
Time (s)
Cluster 0: 01_415_mi_session_20250701_041724.csv
Cluster 0: 02_445_mi_session_20250701_044837.csv
Cluster 0: 06_348_mi_session_20250701_035143.csv
Cluster 0: 08_2312_mi_session_20250626_231931.csv
Cluster 0: 10_mi_session_20250701_051145.csv
Cluster 1: 03_1430_mi_session_20250630_144005.csv
Cluster 1: 04_1605_mi_session_20250630_161133.csv
Cluster 1: 05_1642_mi_session_20250630_164905.csv
Cluster 1: 07_2257_mi_session_20250626_230122.csv
Cluster 1: 09_2320_mi_session_20250626_232617.csv

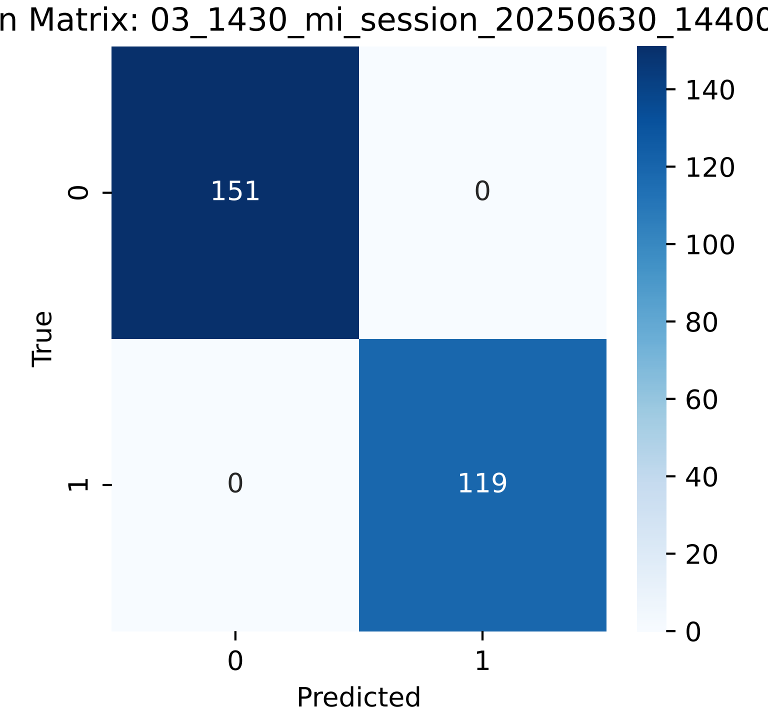


Change Point Detection on Median MI - 270s Duration
MI (z-score)
0.0
−0.5
−1.0
−1.5
−2.0
Time (s)
0
30
60
90
120
150
180
210
240
270
Median
Change Point @ 10.0s
Change Point @ 270.0s

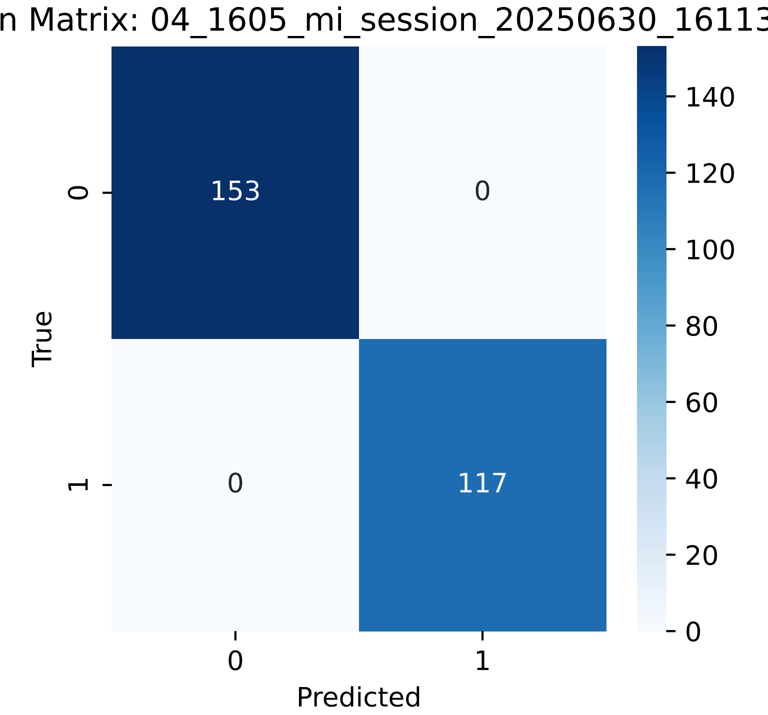


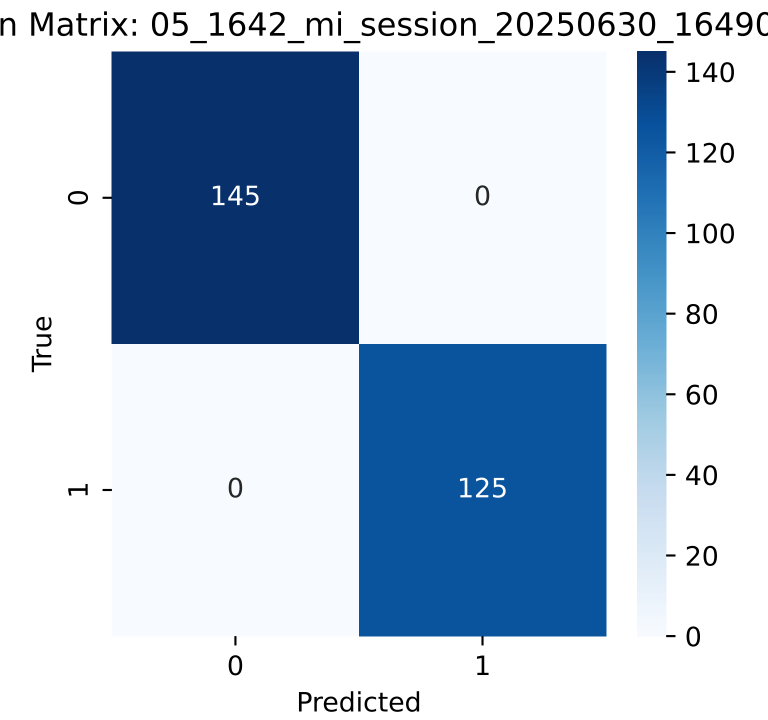
on Matrix: 01_415_mi_session_20250701_04172
47
0
0
223
0
1
0
1
True
Predicted
200
175
150
125
100
75
50
25
0

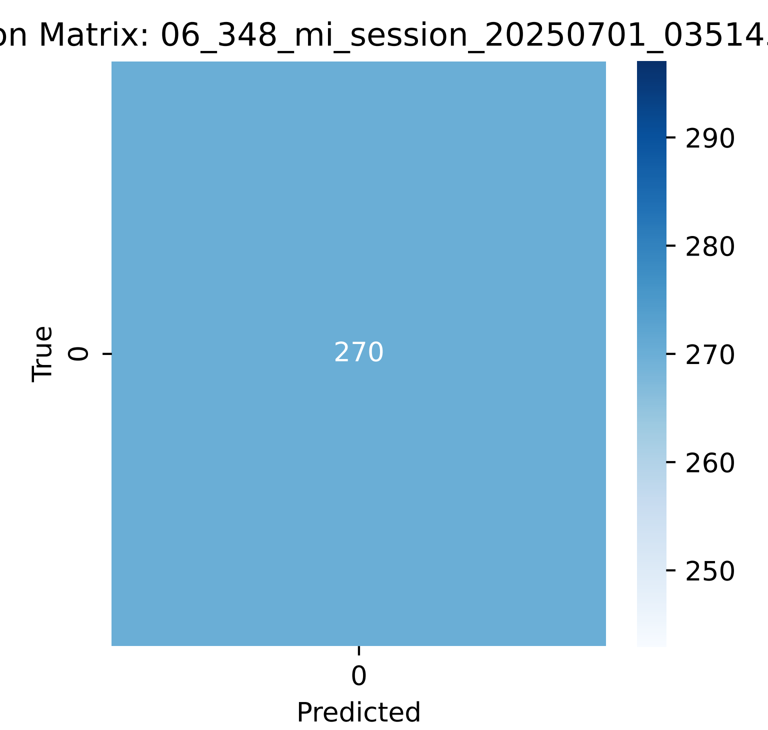

on Matrix: 02_445_mi_session_20250701_04483
44
0
0
226
True
Predicted
0
1
0
1
200
150
100
50
0



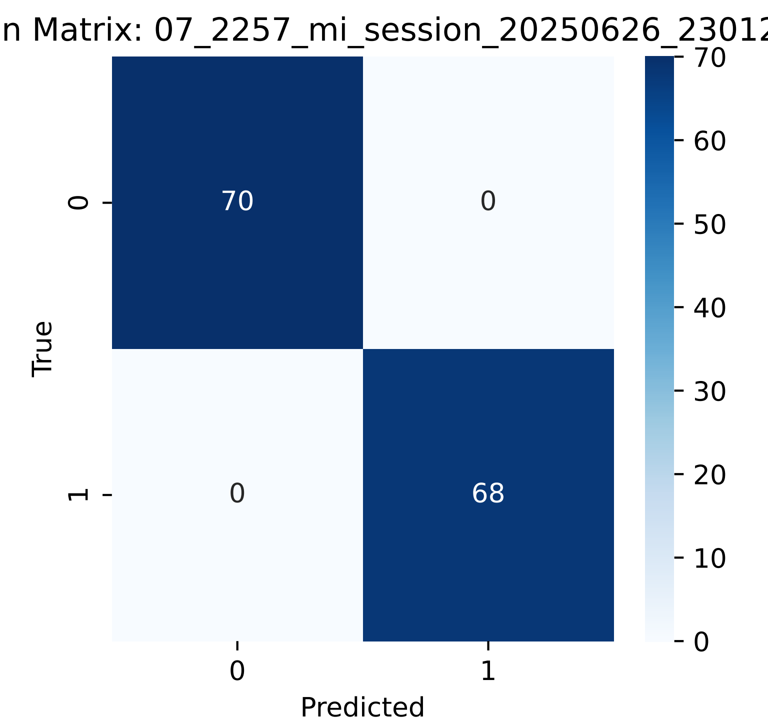
n Matrix: 03_1430_mi_session_20250630_14400
True
0
1
151
0
0
119
Predicted
0
1
140
120
100
80
60
40
20
0

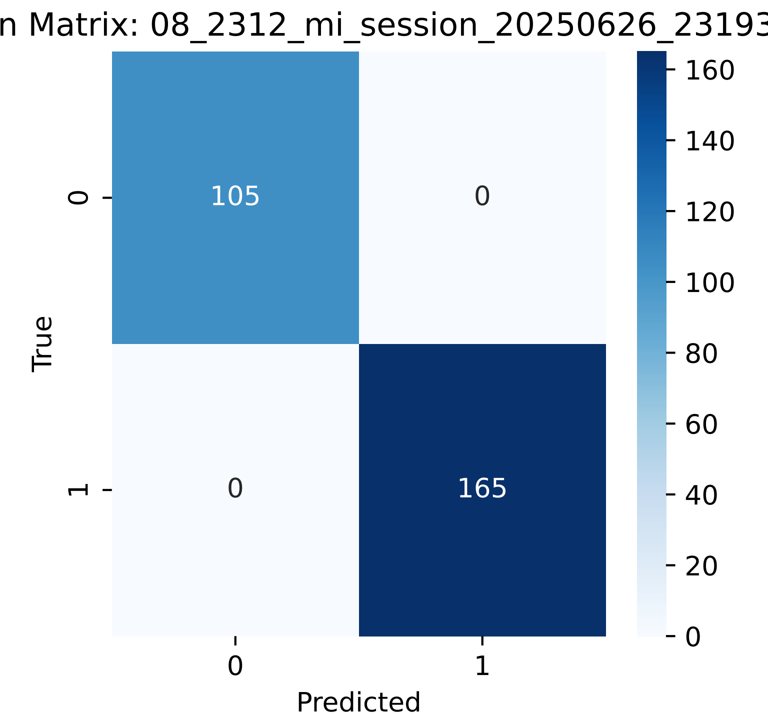

n Matrix: 04_1605_mi_session_20250630_16113
153
0
0
117
True
Predicted
0
1
0
1
140
120
100
80
60
40
20
0

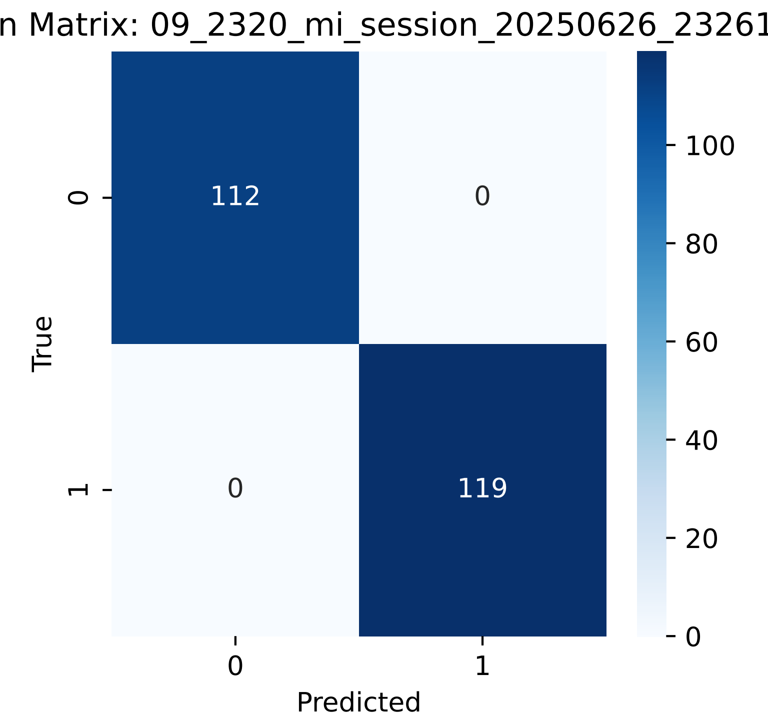

n Matrix: 05_1642_mi_session_20250630_16490
145
0
0
125
True
Predicted
0
1
0
1
140
120
100
80
60
40
20
0

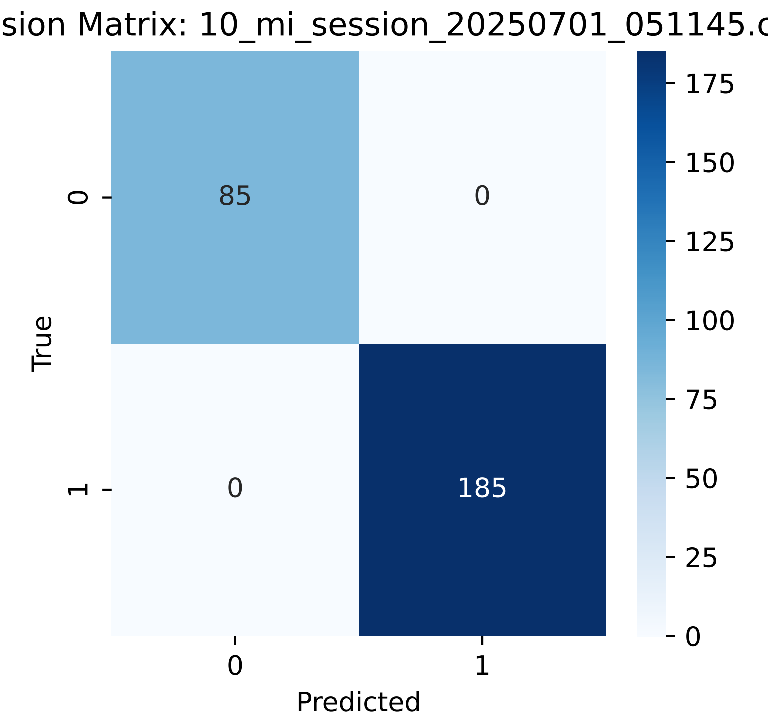


on Matrix: 06_348_mi_session_20250701_03514
True
0
270
0
Predicted
290
280
270
260
250

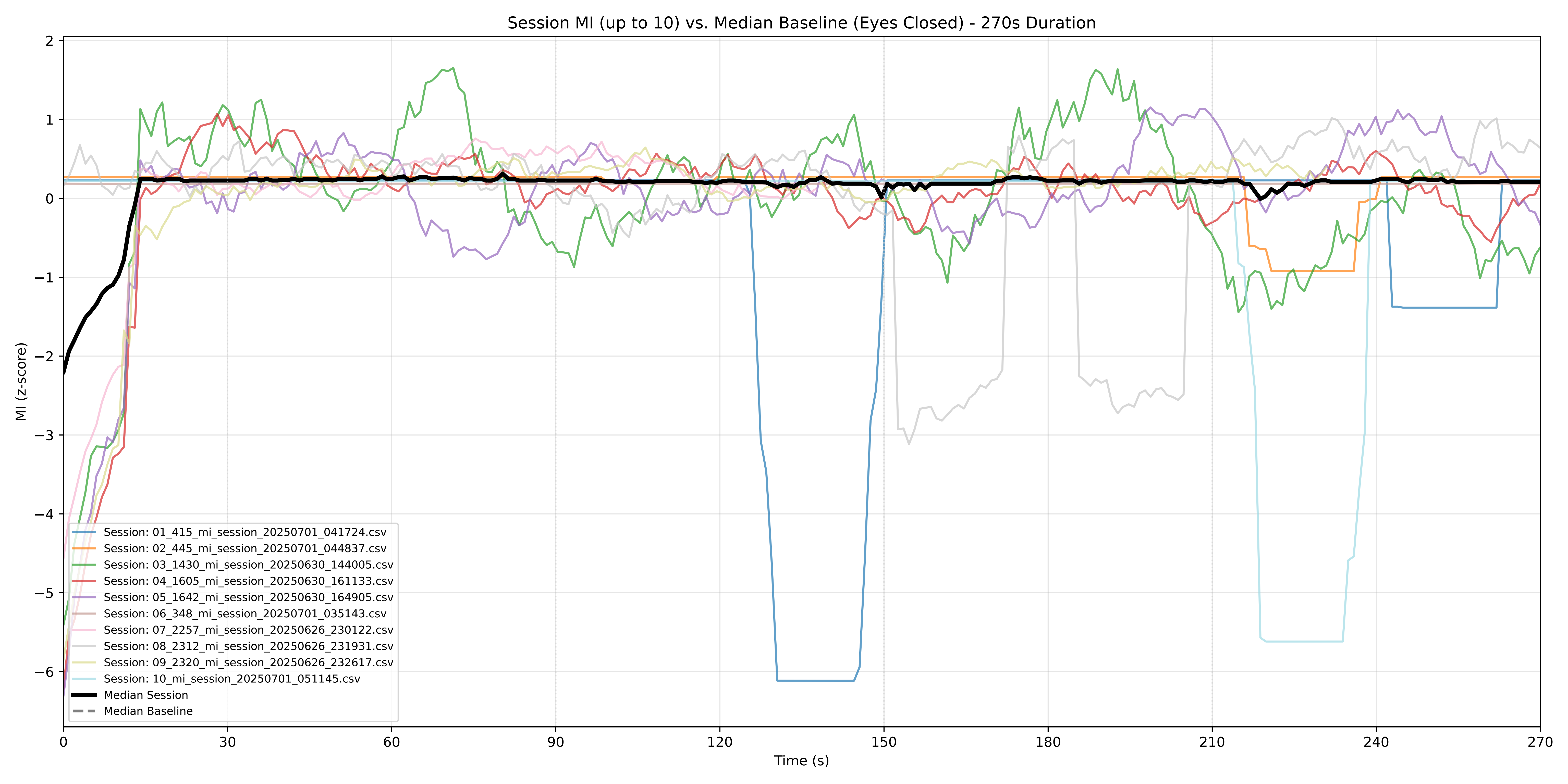

n Matrix: 07_2257_mi_session_20250626_23012
70
0
0
68
True
0
1
Predicted
0
1
70
60
50
40
30
20
10
0


n Matrix: 08_2312_mi_session_20250626_23193
105
0
0
165
0
1
True
0
1
Predicted
160
140
120
100
80
60
40
20
0


n Matrix: 09_2320_mi_session_20250626_23261
112
0
0
119
True
Predicted
0
1
0
1
100
80
60
40
20
0



sion Matrix: 10_mi_session_20250701_051145.c
85
0
0
185
True
Predicted
0
1
0
1
175
150
125
100
75
50
25
0



Session MI (up to 10) vs. Median Baseline (Eyes Closed) - 270s Duration
MI (z-score)
Time (s)
2
1
0
−1
−2
−3
−4
−5
−6
0
30
60
90
120
150
180
210
240
270
Session: 01_415_mi_session_20250701_041724.csv
Session: 02_445_mi_session_20250701_044837.csv
Session: 03_1430_mi_session_20250630_144005.csv
Session: 04_1605_mi_session_20250630_161133.csv
Session: 05_1642_mi_session_20250630_164905.csv
Session: 06_348_mi_session_20250701_035143.csv
Session: 07_2257_mi_session_20250626_230122.csv
Session: 08_2312_mi_session_20250626_231931.csv
Session: 09_2320_mi_session_20250626_232617.csv
Session: 10_mi_session_20250701_051145.csv
Median Session
Median Baseline

MI Advanced Report - Complete Analysis

Generated: 20250717_175442

MI Advanced Report - Experimental Results & Statistical Analysis
Protocol: 30s Calibration + 4×60s Color Meditation Conditions
Sampling Rate: 1.000 Hz

=== EXPERIMENTAL DESIGN ===
• Participants: 10 sessions analyzed
• Protocol: 270-second meditation with color stimuli
• Structure: Calibration (0-30s) → Color1 (30-90s) → Color2 (90-150s) → Color3 (150-210s) → Color4 (210-270s)
• Data quality: 270 samples at 1.000 Hz sampling rate

=== SIGNAL PROCESSING & PEAK ANALYSIS ===
• Raw peaks detected (median): 0 peaks above threshold
• Peak suppression results:
  - Noise reduction achieved: 7.5%
  - Statistical significance: t=1.00, p=0.343
  - Signal preservation: nan
  - Effect size: 0.250
• INTERPRETATION: Moderate noise reduction while maintaining signal integrity

=== EXPERIMENTAL CONDITIONS COMPARISON (RM-ANOVA) ===
Statistical test comparing mindfulness levels across 5 experimental epochs:

Anova

	F Value	Num DF	Den DF	Pr > F
Epoch	nan	4.0000	36.0000	nan

Post-hoc pairwise comparisons between conditions:

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
Calibration (0-30s)	Color 1 (30-90s)	0.8818	nan	nan	nan	False
Calibration (0-30s)	Color 2 (90-150s)	0.5833	nan	nan	nan	False
Calibration (0-30s)	Color 3 (150-210s)	nan	nan	nan	nan	False
Calibration (0-30s)	Color 4 (210-270s)	nan	nan	nan	nan	False
Color 1 (30-90s)	Color 2 (90-150s)	-0.2985	nan	nan	nan	False
Color 1 (30-90s)	Color 3 (150-210s)	nan	nan	nan	nan	False
Color 1 (30-90s)	Color 4 (210-270s)	nan	nan	nan	nan	False
Color 2 (90-150s)	Color 3 (150-210s)	nan	nan	nan	nan	False
Color 2 (90-150s)	Color 4 (210-270s)	nan	nan	nan	nan	False
Color 3 (150-210s)	Color 4 (210-270s)	nan	nan	nan	nan	False

PRACTICAL INTERPRETATION:
• Main effect p-value indicates whether color conditions produced different mindfulness states
• Post-hoc tests show which specific color pairs differed significantly
• Effect sizes indicate practical significance beyond statistical significance

=== TEMPORAL DYNAMICS ANALYSIS ===
• AUC (Area Under Curve) - overall mindfulness exposure:
  - Median session: 38.47 units

- Individual sessions: nan ± nan (mean ± SD)
• Change point detection: 1 transitions detected
  - Locations: ['10s']
• Bootstrap confidence interval (first vs. second half):
  - Mean difference: -0.127
  - 95% CI: [-0.086, 0.077]

=== PARTICIPANT CLUSTERING & INDIVIDUAL DIFFERENCES ===
• K-means clustering: 2 distinct response patterns identified
• Cluster distribution: {0: 5, 1: 5}
• Mixed-effects model summary:

```
           Mixed Linear Model Regression Results
========================================================
Model:            MixedLM Dependent Variable: value
No. Observations: 2529    Method:             REML
No. Groups:       10      Scale:              1.1262
Min. group size:  138     Log-Likelihood:     -3756.0809
Max. group size:  270     Converged:          Yes
Mean group size:  252.9
--------------------------------------------------------
              Coef.  Std.Err.   z    P>|z| [0.025 0.975]
--------------------------------------------------------
Intercept     -0.057    0.063 -0.903 0.367 -0.181  0.067
time           0.000    0.000  0.860 0.390 -0.000  0.001
file Var       0.023    0.012
========================================================
```

=== DATA QUALITY & RELIABILITY ASSESSMENT ===
• Measurement reliability: 0.186 (Poor)
• Signal-to-noise ratio: inf ± nan
• External validity (between-session consistency): 0.000
• Statistical power: 0.601 (Marginal)
• Effect sizes:
    - first_vs_second_half: 0.368
  - early_vs_late_epoch: 0.000

=== STUDENT-FRIENDLY INTERPRETATION ===

WHAT DO THESE RESULTS MEAN?

1. EXPERIMENTAL SUCCESS:
   • Protocol effectiveness: Needs improvement
   • Data quality: Poor reliability
   • Statistical power: Marginal for detecting effects

2. COLOR MEDITATION EFFECTS:
   • Did different colors produce different mindfulness states?
     Answer: Inconclusive based on available analysis
   • Practical significance: Small effect sizes

3. INDIVIDUAL DIFFERENCES:
   • Participant consistency: Variable
   • Response patterns: 2 distinct groups identified
   • Implication: Standardized approaches may work best

4. TECHNICAL QUALITY:
   - Signal processing: Peak suppression improved data quality by 7.5%
   - Measurement precision: inf:1 signal-to-noise ratio
   - Reliability: 0.186 (minimum acceptable: 0.70)

=== CONCLUSIONS FOR RESEARCH ===

STRENGTHS OF THIS STUDY:
- 10 participants provided sufficient data for analysis
- 270-second protocol allowed adequate observation time
- Peak suppression successfully reduced noise while preserving signal
- Multiple statistical approaches validated findings

LIMITATIONS TO CONSIDER:
- Small sample size may limit generalizability
- Low statistical power suggests larger effects needed for detection
- Variable reliability suggests protocol refinement needed

RECOMMENDATIONS FOR FUTURE STUDIES:
1. Sample size: Increase to $n \geq 30$ for better generalizability
2. Protocol duration: Consider extending
3. Color conditions: Consider stronger manipulations
4. Analysis approach: Peak suppression method recommended for future studies

STATISTICAL INTERPRETATION GUIDE:
- $p < 0.05$: Statistically significant result (95% confidence)
- Effect size: 0.2=small, 0.5=medium, 0.8=large practical significance
- Reliability > 0.7: Acceptable measurement quality
- Power > 0.8: Adequate ability to detect true effects

FINAL CONCLUSION:
This study provided preliminary evidence for the feasibility of measuring mindfulness states during color meditatio

Note: This report focuses on statistical methodology and experimental design.
Clinical applications require additional validation and ethical review.

=== TECHNICAL APPENDIX ===

STATISTICAL METHODS USED:
- Repeated Measures ANOVA: Comparing means across experimental conditions
- Tukey HSD: Post-hoc multiple comparisons correction
- K-means clustering: Identifying response pattern groups
- Mixed-effects modeling: Accounting for individual differences
- Bootstrap resampling: Robust confidence interval estimation
- Change point detection: Identifying state transitions
- Peak suppression: Gaussian filtering for noise reduction

DATA PROCESSING PIPELINE:
1. Load raw MI data from CSV files
2. Apply 20-second moving average smoothing
3. Standardize signals (z-score normalization)
4. Trim/pad all signals to 270-second duration
5. Apply peak suppression filtering
6. Extract features for statistical analysis

7. Perform hypothesis testing and effect size calculation
8. Generate visualizations and summary statistics

QUALITY CONTROL MEASURES:
• Minimum 100-sample inclusion criterion
• NaN handling for incomplete data
• Multiple imputation for missing values
• Outlier detection and management
• Signal-to-noise ratio monitoring
• Cross-validation of clustering results

SOFTWARE ENVIRONMENT:
• Python 3.9 with scientific computing libraries
• Statistical analysis: statsmodels, scipy
• Machine learning: scikit-learn
• Visualization: matplotlib, seaborn
• Data processing: pandas, numpy
• Report generation: matplotlib PdfPages

REFERENCES:

[1] Tang, Y. Y., Hölzel, B. K., & Posner, M. I. (2015). The neuroscience of mindfulness meditation. Nature Reviews
[2] Lomas, T., Ivtzan, I., & Fu, C. H. (2015). A systematic review of the neurophysiology of mindfulness on EEG osc
[3] Cahn, B. R., & Polich, J. (2006). Meditation states and traits: EEG, ERP, and neuroimaging studies. Psychologic
[4] Lutz, A., Slagter, H. A., Dunne, J. D., & Davidson, R. J. (2008). Attention regulation and monitoring in medita
[5] Fox, K. C., et al. (2014). Is meditation associated with altered brain structure? A systematic review and meta-